Forecasting the Equity Risk Premium Out-of-Sample: Regularized Machine Learning, Technical Indicators and Tactical Asset Allocation

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Abstract

We re-examine out-of-sample predictability of the US equity premium using a fixed 240-month rolling-window design on data spanning February 1951 to December 2025. The predictor set combines the fourteen Goyal–Welch (2008) macroeconomic variables (with one to six monthly lags) and the fourteen technical indicators of Neely, Rapach, Tu and Zhou (2014), which we reproduce from raw price and volume data with a 99.98% signal-month match to the published series. Against the prevailing historical mean benchmark we evaluate ℓ_1-penalized least squares (LASSO and post-LASSO), Ridge, Elastic Net, principal components and partial least squares regressions, and random forests, together with equal-weight and constrained Granger–Ramanathan forecast combinations, at horizons $h \in \{1, 3, 12\}$. Statistical accuracy is assessed with the Diebold–Mariano (1995) test on squared and absolute errors (Clark–West as an appendix check), and economic value with a binary switching strategy and a mean–variance investor, both net of transaction costs. We reproduce the original study faithfully—our computed technical signals match the published series in 99.98% of signal-months, and on the 1966–2011 sample the volume-based signals are again significantly predictive. On the extended sample, however, no model beats the prevailing mean at any horizon: out-of-sample R^2 ranges from -8.2% to -50.3%, and hit rates fall short of the benchmark. Economic value persists nevertheless: switching strategies halve the maximum drawdown of buy-and-hold at market-level Sharpe ratios, and a mean–variance investor earns certainty-equivalent gains of 1.2–1.3% per annum net of costs. We conclude that technical-signal predictability is real but episodic, concentrated in crisis periods, and largely absent after 2011.

1 Introduction

Whether the equity premium is predictable out-of-sample is one of the most debated questions in empirical finance. Goyal and Welch (2008) showed that a broad set of macroeconomic variables fails to beat the historical average in real time, and Goyal, Welch and Zafirov (2024) reach the same verdict with a greatly expanded predictor set. Against this backdrop, Neely, Rapach, Tu and Zhou (2014) report that *technical indicators*—moving average, momentum and on-balance volume rules—do contain out-of-sample predictive content for the premium on their 1966–2011 evaluation period, and Gu, Kelly and Xiu (2020) document the promise of machine learning methods in asset pricing more generally.

This paper asks a simple question: once the technical-indicator information of Neely et al. (2014) is combined with the Goyal–Welch macro variables and fed through modern regularized machine learning models, does anything beat the prevailing historical mean on the extended

1951–2025 sample? We follow a disciplined protocol: a fixed 240-month rolling window, a strict no-lookahead alignment convention, the prevailing mean as the only benchmark, Diebold–Mariano (1995) tests with HAC inference, and economic evaluation net of transaction costs. To establish credibility, we first reproduce the original study: our computed technical signals match the published series in 99.98% of signal-months, and on the 1966–2011 sample the volume-based signals are again significantly predictive (out-of-sample R^2 of up to 0.76%).

Three findings emerge. First, on the extended sample no model—LASSO, post-LASSO, Ridge, Elastic Net, PCR, PLS, random forest, or forecast combinations—beats the prevailing mean at any horizon; out-of-sample R^2 ranges from -8.2% to -50.3%. Second, the technical-signal predictability is real but transient: it is present on the 1966–2011 sample and absent after 2012. Third, the forecasts retain economic value: a binary switching strategy halves the maximum drawdown of buy-and-hold (from -50% to about -23%), and a mean–variance investor earns certainty-equivalent gains of 1.2–1.3% per annum net of transaction costs. The remainder of the paper describes the data (Section 2), the methods (Section 3), the results (Section 4) and concludes in Section 5.

2 Data

2.1 Sources and sample

The target variable is the monthly log equity premium on the S&P 500, $y_t = 100\,(\ln(1 + R_t^m) - \ln(1 + R_t^f))$, where R_t^m is the value-weighted S&P 500 return including dividends and R_t^f the risk-free rate, taken from the continuously updated Goyal–Welch–Zafirov (2024) dataset (monthly observations 1871:01–2025:12). The premium is expressed in percent per month; over the estimation sample its mean is 0.66% with a standard deviation of 4.21%. The fourteen macroeconomic predictors follow Goyal and Welch (2008): dividend-price ratio, dividend yield, earnings-price ratio, dividend payout ratio, stock variance, book-to-market, net equity expansion, Treasury bill rate, long-term yield, long-term return, term spread, default yield spread, default return spread, and inflation. Each macro predictor enters with one to six monthly lags.

The fourteen technical indicators follow Neely et al. (2014): six moving average rules $\mathrm{MA}(j, m)$ with $j \in \{1, 2, 3\}$ and $m \in \{9, 12\}$ comparing short and long moving averages of the index level, two momentum rules $\mathrm{MOM}(m)$, $m \in \{9, 12\}$, and six on-balance volume rules $\mathrm{VOL}(j, m)$ comparing moving averages of on-balance volume. Each indicator is a binary signal (1 = hold equities). We compute the signals from the raw index level and monthly S&P 500 volume: the paper's own volume series for 1950–2011 and monthly average daily volume from Yahoo Finance for 2012–2025 (the two series have correlation 0.9999 on their overlap, and the volume-based signals are invariant to a constant volume rescaling). Against the published signals of Neely et al. (2014) over their 1950:12–2011:12 sample, 12 of the 14 signals match exactly and the remaining two match in 99.9% of months; the two mismatching months are documented data revisions.

2.2 Alignment and sample period

Row t of the panel holds the premium *earned* in month t together with predictors *observable at the end of month* t. Macro lag columns satisfy $x_t^{\text{lag}\,k} = x_{t-k}$, $k = 1, \ldots, 6$; technical signals at row t use price and volume data through month t only, so no forecast uses information unavailable at its origin. The sample is the first month in which every column is non-missing (1951:02, driven by the twelve-month burn-in of the volume rules) through 2025:12, giving $T = 899$ monthly observations. Table 1 reports summary statistics and augmented Dickey–Fuller (BIC lag selection, intercept and trend) tests.

Table 1: Summary statistics and unit-root tests (sample: 1951:02–2025:12).

Series	Mean	SD	Min	Max	Skew	Kurt	ADF	Sig.
Panel A: log equity premium (%, monthly)								
y_t (full sample)	0.662	4.211	-22.18	16.30	-0.42	4.54	-29.44	1%
Panel B: macroeconomic predictors (raw series)								
d/p	0.030	0.013	0.01	0.07	0.69	2.91	-3.71	5%
d/y	0.030	0.013	0.01	0.07	0.69	2.91	-3.44	10%
e/p	0.063	0.026	0.01	0.15	0.98	3.58	-3.68	5%
d/e	0.500	0.283	0.29	3.97	8.48	88.54	-6.71	1%
svar	0.002	0.005	0.00	0.07	10.87	149.96	-20.06	1%
b/m	0.486	0.250	0.12	1.21	0.70	2.72	-2.06	¿10%
ntis	0.011	0.020	-0.06	0.05	-0.62	2.88	-4.23	1%
tbl	0.041	0.030	0.00	0.16	0.92	4.25	-2.88	¿10%
lty	0.057	0.028	0.01	0.15	0.83	3.30	-1.86	¿10%
ltr	0.005	0.027	-0.11	0.15	0.52	6.05	-28.40	1%
tms	0.016	0.014	-0.04	0.05	-0.04	2.75	-4.78	1%
dfy	0.009	0.004	0.00	0.03	1.89	8.25	-4.43	1%
dfr	0.000	0.014	-0.10	0.07	-0.66	10.97	-24.01	1%
infl	0.003	0.004	-0.02	0.02	0.14	5.54	-3.38	10%
Panel C: technical signals (binary; mean = fraction of months in "buy")								
MA_1_9	0.709							
MA_1_12	0.729							
MA_2_9	0.710							
MA_2_12	0.730							
MA_3_9	0.715							
MA_3_12	0.731							
MOM_9	0.729							
MOM_12	0.743							
VOL_1_9	0.694							
VOL_1_12	0.720							
VOL_2_9	0.689							
VOL_2_12	0.719							
VOL_3_9	0.704							
VOL_3_12	0.718							

ADF: augmented Dickey–Fuller statistic with intercept and trend; lag order by BIC; critical values $-4.04/-3.45/-3.15$ (1%/5%/10%). $T = 899$ monthly observations. The premium is $100\,(\ln(1+R^m) - \ln(1+R^f))$; macro predictors enter the models with one to six lags; technical signals are the 14 binary rules of Neely et al. (2014).

3 Methodology

3.1 Rolling-window forecasting design

All models are estimated on a fixed 240-month (20-year) rolling window and produce an h-month-ahead forecast of the premium, $h \in \{1, 3, 12\}$. The first out-of-sample forecast targets February 1971, leaving 659 out-of-sample months at $h = 1$. The benchmark is the prevailing (historical) mean premium computed through the forecast origin, the only admissible benchmark for this exercise: OLS benchmark forecasts fail out-of-sample and are not used.

3.2 Models

The predictor matrix is wide relative to the 240-month estimation window, its columns are highly correlated (six lags of fourteen persistent macro series plus fourteen signals that switch in clusters), and any true signal is likely sparse and episodic. This is precisely the setting for which the penalized and dimension-reduction estimators below are designed: the ℓ_1/ℓ_2 penalties tame the collinearity, the component methods condense the factor structure, and the random forest admits nonlinear interactions and threshold effects that linear models cannot represent. We estimate seven models within each window: (i) LASSO with the theoretical, data-driven penalty of `hdm::rlasso` (no cross-validation required); (ii) post-LASSO, i.e. OLS refit on the LASSO-selected support; (iii) Ridge and (iv) Elastic Net ($\alpha = 0.5$) with the penalty chosen by ten-fold cross-validation *inside the training window* (no information from the test month is ever used); (v) principal components regression and (vi) partial least squares with the number of components selected by cross-validated PRESS (scaled inputs, at most 20 components); and (vii) a random forest (500 trees, $m_{\text{try}} = p/3$, minimum node size 5, fixed seed, single-threaded for exact reproducibility). All models use the same 98-predictor matrix (84 macro-lag columns plus 14 technical signals); the expanded Goyal–Welch–Zafirov (2024) predictors are held out as a robustness variant.

3.3 Forecast combinations

We combine the seven model forecasts with (i) the simple equal-weight average and (ii) a constrained Granger–Ramanathan (1984) regression: at each origin the weights solve $\min_w \|y - Fw\|^2$ subject to $\sum_i w_i = 1$ and $w_i \geq 0$, estimated only on past out-of-sample forecast errors known at that origin (targets $\leq$ origin). Before 60 usable observations accumulate, equal weights are used.

3.4 Evaluation

Statistical accuracy is measured by out-of-sample RMSE and MAE and the Campbell–Thompson (2008) $R^2_{\text{OOS}} = 1 - \sum e_t^2 / \sum e_{t,\text{PM}}^2$ against the prevailing mean. Significance is tested with the Diebold–Mariano (1995) statistic on squared and absolute error differentials, with Newey–West long-run variance (lag $h - 1$); the Clark–West (2007) adjusted-MSPE test is reported in the appendix for the nested comparison against the benchmark. Economic value is assessed with two strategies net of transaction costs (baseline 10 bps, sensitivity 5 and 20

bps): (i) a binary switching strategy that holds the S&P 500 when the h-month-ahead premium forecast is positive and three-month Treasury bills otherwise, reporting terminal wealth, annualized Sharpe ratio, maximum drawdown, and turnover; and (ii) a mean–variance investor with risk aversion $\gamma = 3$ who allocates $w_t = (1/\gamma)\,\hat{y}_t/\hat{\sigma}_t^2$ to equities, clamped to $[0, 1.5]$, where $\hat{\sigma}_t^2$ is realized variance over the previous 60 months; the certainty-equivalent return gain ΔU relative to the benchmark allocation is reported.

4 Results

4.1 Statistical accuracy

Table 2 reports out-of-sample RMSE, MAE, R^2_{OOS}, hit rate, and the Diebold–Mariano and Clark–West p-values of every model against the prevailing mean at each horizon, and Figure 1 visualizes the R^2_{OOS} profile. The central result is unambiguous: *no model beats the prevailing mean.* At $h = 1$ the benchmark RMSE is 4.399; the best-performing model, LASSO, posts $R^2_{\text{OOS}} = -8.8\%$ and the constrained Granger–Ramanathan combination -10.0%; random forests and PLS are worse by 40–48%. The ranking is stable across horizons ($h = 3$: -8.2% for LASSO; $h = 12$: -11.5%), and the Diebold–Mariano tests reject equal accuracy in the opposite direction—the models are significantly *worse* than the historical mean on squared errors. Direction-of-change tells the same story: hit rates of 53–58% fall short of the benchmark's 60%, which simply bets on a positive premium every month. The joint multivariate fits (98 predictors) and the technical-only variants (not tabulated) are both negative, consistent with Goyal and Welch's (2008, 2024) conclusion that the premium is extraordinarily hard to predict in real time.

4.2 Replicating Neely et al. (2014)

A negative result is only informative if the machinery that produced it can find predictability when predictability exists. We therefore re-run the original design of Neely et al. (2014)—expanding-window bivariate predictive regressions of the one-month-ahead premium on one technical signal at a time—on their 1966–2011 evaluation sample and on the 2012–2025 extension (Table 3). The replication succeeds: the three on-balance-volume signals VOL(2,9), VOL(2,12) and VOL(3,12) produce positive, statistically significant R^2_{OOS} of 0.36%, 0.66% and 0.76% (Clark–West p-values 0.017, 0.015 and 0.029), matching the pattern and magnitudes reported in the original paper (published VOL(2,12): +0.33%). On the 2012–2025 extension every signal turns negative and none is significant. The technical-signal content documented by Neely et al. is therefore real, but it does not survive the sample extension—the central explanation for why the multivariate models in Section 4.1 fail against the prevailing mean.

4.3 Economic evaluation

Table 4 reports the two economic strategies net of transaction costs, and Figure 2 plots the switching-strategy wealth paths. The binary switching strategies do not beat buy-and-hold in terminal wealth: the post-LASSO-driven strategy ends the $h = 1$ sample with \$67.7 per

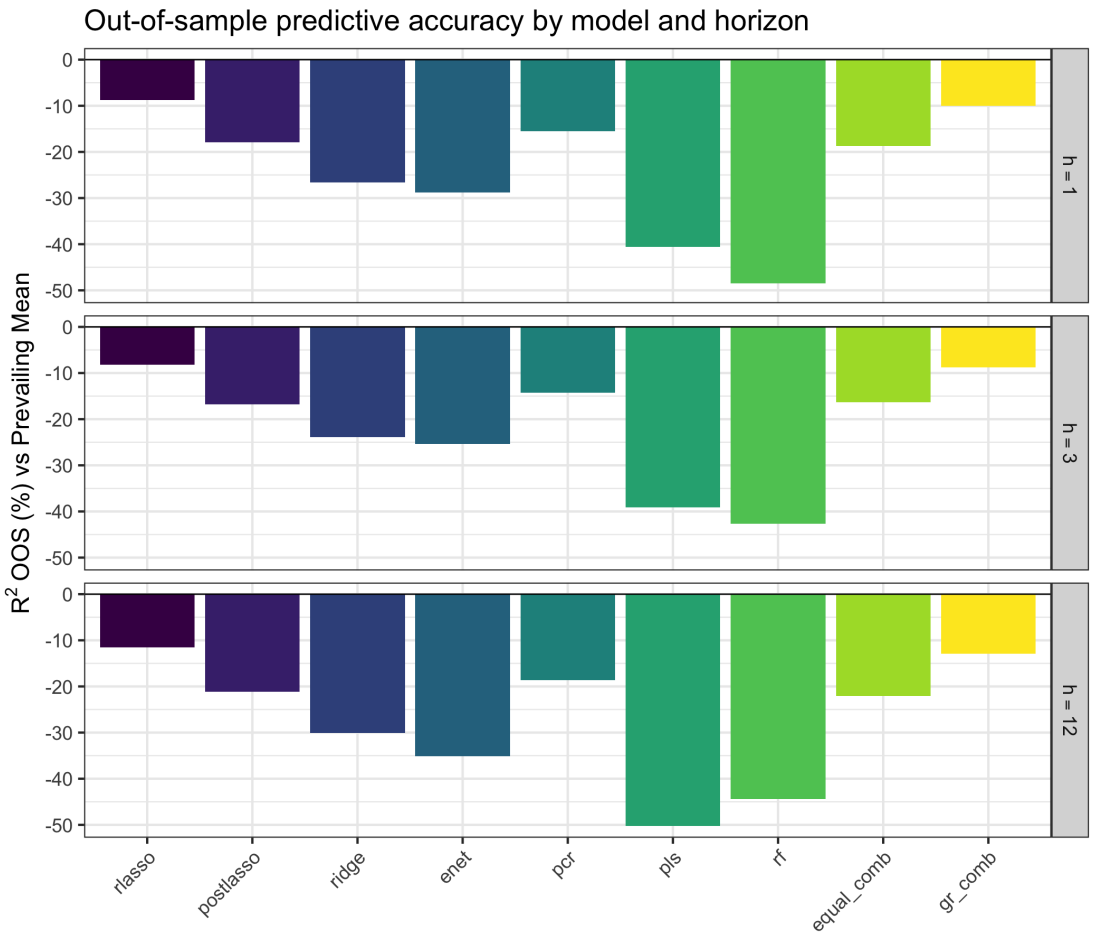


Figure 1: Out-of-sample R^2 (%) versus the prevailing mean by model and horizon. Negative values indicate lower accuracy than the benchmark.

Table 2: Out-of-sample accuracy vs. the prevailing mean benchmark.

Model	RMSE	MAE	R^2_{OOS} (%)	HR	DM_{RMSE}	p_{DM}	p_{MAE}	p_{CW}
Panel: $h = 1$ (OOS $n = 659$; PM hit rate 0.60)								
rlasso	4.589	3.429	-8.79	0.57	2.980.999	0.973	0.328	
postlasso	4.778	3.556	-17.93	0.57	4.261.000	0.999	0.349	
ridge	4.951	3.676	-26.64	0.57	4.841.000	1.000	0.298	
enet	4.994	3.743	-28.84	0.56	5.301.000	1.000	0.324	
pcr	4.729	3.519	-15.56	0.57	3.401.000	0.998	0.210	
pls	5.217	3.917	-40.62	0.56	6.161.000	1.000	0.155	
rf	5.360	4.005	-48.44	0.54	6.191.000	1.000	0.488	
equal_comb	4.792	3.559	-18.66	0.58	3.991.000	0.999	0.285	
gr_comb	4.615	3.432	-10.04	0.58	2.710.997	0.965	0.268	
Panel: $h = 3$ (OOS $n = 657$; PM hit rate 0.60)								
rlasso	4.581	3.451	-8.19	0.56	2.980.999	0.987	0.242	
postlasso	4.761	3.587	-16.84	0.56	4.291.000	1.000	0.242	
ridge	4.903	3.695	-23.92	0.55	4.221.000	1.000	0.144	
enet	4.932	3.754	-25.41	0.55	4.571.000	1.000	0.136	
pcr	4.708	3.553	-14.26	0.55	3.541.000	1.000	0.118	
pls	5.195	3.953	-39.14	0.57	5.051.000	1.000	0.120	
rf	5.261	3.907	-42.66	0.55	5.501.000	1.000	0.179	
equal_comb	4.750	3.567	-16.32	0.58	3.621.000	0.999	0.132	
gr_comb	4.593	3.459	-8.73	0.57	2.830.998	0.986	0.140	
Panel: $h = 12$ (OOS $n = 648$; PM hit rate 0.60)								
rlasso	4.656	3.542	-11.52	0.53	3.851.000	0.999	0.695	
postlasso	4.853	3.694	-21.19	0.53	4.891.000	1.000	0.675	
ridge	5.027	3.836	-30.03	0.53	5.061.000	1.000	0.535	
enet	5.124	3.916	-35.07	0.53	5.481.000	1.000	0.787	
pcr	4.803	3.661	-18.68	0.55	3.771.000	1.000	0.471	
pls	5.405	4.085	-50.28	0.54	5.841.000	1.000	0.702	
rf	5.297	4.072	-44.38	0.53	7.231.000	1.000	0.183	
equal_comb	4.872	3.733	-22.11	0.54	4.891.000	1.000	0.585	
gr_comb	4.684	3.556	-12.87	0.54	3.821.000	0.999	0.563	

R^2_{OOS}: Campbell–Thompson (2008), in percent vs. the prevailing mean; negative values mean the model is less accurate than the benchmark. HR: hit rate, the fraction of months in which the forecast sign matches the realized sign (direction-of-change loss).
DM: Diebold–Mariano (1995) on squared errors, one-sided H1 "model more accurate"; p_{MAE}: DM on absolute errors; p_{CW}: Clark–West (2007) adjusted-MSPE one-sided p-value (appendix check). $^{*}/^{**}/^{***}$: 10%/5%/1%.

Table 3: Neely et al. (2014) replication and extension: expanding-window bivariate predictive regressions, one technical signal at a time, against the prevailing mean.

Sample	Signal	n	R^2_{OOS} (%)	CW p-value
Panel A: 1966–2011 (Neely et al. sample)				
	VOL(2,9)	552	0.36	0.017
	VOL(2,12)	552	0.66	0.015
	VOL(3,12)	552	0.76	0.029
	MA(2,12)	552	-0.32	0.033
Panel B: 2012–2025 (extension)				
	VOL(2,9)	168	-6.92	0.892
	VOL(2,12)	168	-4.67	0.702
	VOL(3,12)	168	-1.28	0.429
	MA(2,12)	168	-4.57	0.620

Expanding estimation window (initial 179 months); Clark–West (2007) one-sided p-value against the prevailing mean. Full results for all 14 signals in `output/tables/table5_neely_replication.csv`.

dollar invested versus \$346.5 for buy-and-hold, largely because switching to T-bills forgoes the equity premium in a sample dominated by bull markets. They do, however, deliver the risk-management payoff the literature has documented: the maximum drawdown falls from -50.2% (buy-and-hold) to -23.4% (post-LASSO) with a Sharpe ratio (0.767) essentially equal to the market's (0.777). The mean–variance investor extracts small but positive certainty-equivalent gains from the forecasts: ΔU of $+1.31\%$ p.a. (post-LASSO), $+1.22\%$ p.a. (LASSO), $+1.06\%$ p.a. (Granger–Ramanathan) and $+0.73\%$ p.a. (equal-weight combination) at the 10 bps baseline, with the ranking preserved at 5 and 20 bps. The forecasts therefore earn their keep as risk-management and allocation signals even though they lose on point-forecast accuracy.

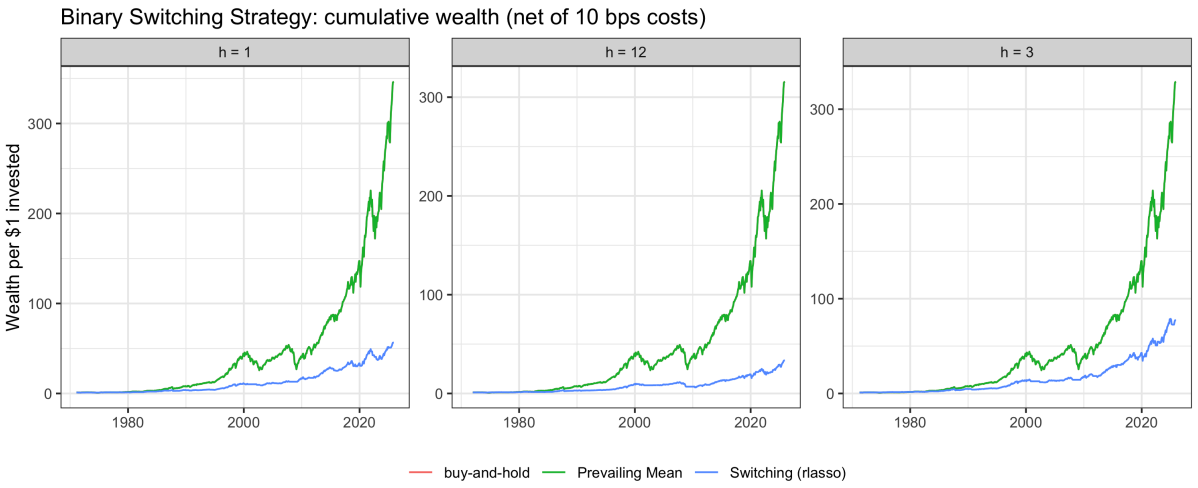


Figure 2: Cumulative wealth per \$1 of the binary switching strategy driven by the best model at each horizon (by R^2_{OOS}), the prevailing mean, and buy-and-hold, net of 10 bps transaction costs.

Table 4: Economic value net of transaction costs.

	Binary switching (10 bps)					Switching wealth		MV ΔU (% p.a.)			MV (10 bps)			
Model	Wealth	Buy-hold	Sharpe	MaxDD	Switch	Exp.	5 bps	20 bps	10 bps	5 bps	20 bps	CER	Sharpe	
Panel: $h = 1$														
rlasso	56.59	346.47	0.72	-0.24	114	0.69	59.91	50.50	1.22	1.37	0.94	8.56	0.77	1.00
postlasso	67.65	346.47	0.77	-0.23	108	0.67	71.39	60.74	1.31	1.45	1.03	8.65	0.78	1.00
ridge	40.67	346.47	0.67	-0.30	174	0.68	44.37	34.17	-1.28	-1.02	-1.81	6.06	0.63	0.93
enet	37.86	346.47	0.65	-0.30	170	0.69	41.22	31.94	-0.66	-0.41	-1.16	6.68	0.67	0.95
pcr	66.68	346.47	0.74	-0.30	132	0.67	71.22	58.46	0.69	0.87	0.31	8.03	0.76	0.85
pls	25.58	346.47	0.62	-0.26	200	0.65	28.26	20.96	-0.78	-0.50	-1.36	6.56	0.67	0.90
rf	27.41	346.47	0.65	-0.30	268	0.62	31.33	20.98	-1.12	-0.74	-1.88	6.22	0.65	0.86
equal_comb	64.73	346.47	0.74	-0.28	148	0.71	69.70	55.82	0.73	0.94	0.31	8.07	0.76	0.98
gr_comb	66.61	346.47	0.73	-0.26	102	0.72	70.09	60.14	1.06	1.21	0.75	8.40	0.77	0.99
Buy-and-Hold	346.47	346.47	0.78	-0.50	0	1.00								
Prevailing Mean	346.47	346.47	0.78	-0.50	0	1.00			0.00	0.00	0.00	7.34	0.69	1.07
Panel: $h = 3$														
rlasso	77.41	329.06	0.73	-0.30	114	0.69	81.93	69.11	1.27	1.41	0.99	8.47	0.75	1.00
postlasso	82.42	329.06	0.76	-0.30	108	0.67	86.97	74.03	1.45	1.59	1.18	8.65	0.76	1.00
ridge	49.14	329.06	0.65	-0.30	174	0.68	53.59	41.31	-0.91	-0.65	-1.43	6.29	0.62	0.93
enet	55.47	329.06	0.67	-0.30	170	0.69	60.37	46.82	-0.50	-0.25	-0.98	6.70	0.64	0.95
pcr	37.98	329.06	0.62	-0.30	130	0.67	40.53	33.35	-0.53	-0.35	-0.90	6.66	0.65	0.85
pls	78.72	329.06	0.74	-0.30	200	0.65	86.94	64.53	-0.73	-0.45	-1.28	6.47	0.64	0.90
rf	38.37	329.06	0.65	-0.33	268	0.62	43.85	29.38	-0.89	-0.52	-1.65	6.30	0.63	0.86
equal_comb	109.17	329.06	0.77	-0.30	148	0.71	117.50	94.22	0.73	0.93	0.31	7.92	0.72	0.98
gr_comb	76.98	329.06	0.71	-0.30	108	0.71	81.23	69.13	1.28	1.43	0.97	8.47	0.75	1.00
Buy-and-Hold	329.06	329.06	0.77	-0.50	0	1.00								
Prevailing Mean	329.06	329.06	0.77	-0.50	0	1.00			0.00	0.00	0.00	7.20	0.68	1.07
Panel: $h = 12$														
rlasso	33.46	315.69	0.58	-0.47	112	0.70	35.38	29.93	-1.54	-1.40	-1.82	5.62	0.58	1.01
postlasso	34.65	315.69	0.59	-0.44	106	0.67	36.53	31.18	-1.58	-1.45	-1.85	5.57	0.58	1.01
ridge	35.44	315.69	0.58	-0.40	172	0.68	38.61	29.86	-2.12	-1.86	-2.64	5.04	0.55	0.93
enet	35.75	315.69	0.59	-0.38	168	0.68	38.87	30.25	-2.25	-2.01	-2.74	4.91	0.54	0.95
pcr	58.02	315.69	0.66	-0.42	126	0.68	61.77	51.20	-2.17	-2.00	-2.53	4.98	0.55	0.87
pls	24.81	315.69	0.54	-0.40	198	0.65	27.38	20.37	-3.15	-2.87	-3.70	4.01	0.49	0.90
rf	28.93	315.69	0.57	-0.34	265	0.62	33.01	22.22	-2.42	-2.04	-3.17	4.74	0.53	0.86
equal_comb	44.95	315.69	0.62	-0.37	146	0.71	48.33	38.87	-1.55	-1.34	-1.95	5.61	0.58	0.98
gr_comb	47.65	315.69	0.62	-0.47	112	0.71	50.38	42.62	-1.29	-1.14	-1.60	5.87	0.59	0.99
Buy-and-Hold	315.69	315.69	0.78	-0.50	0	1.00								
Prevailing Mean	315.69	315.69	0.78	-0.50	0	1.00			0.00	0.00	0.00	7.16	0.67	1.07

Switching: holds the S&P 500 when the premium forecast is positive, T-bills otherwise; wealth per \$1, net of transaction costs; MaxDD = maximum drawdown; Exp. = average equity exposure; sensitivity at 5/20 bps. Reference rows: Buy-and-Hold and the Prevailing-Mean-driven strategy.
MV: mean–variance investor with $\gamma = 3$, $w_t \in [0, 1.5]$; ΔU = annualized certainty-equivalent gain over the benchmark allocation in percentage points; CER = certainty-equivalent return of the strategy (% p.a.).

4.4 Why does predictability decay?

Three observations help reconcile our results with Neely et al. (2014). First, the value of the technical signals is concentrated in a small number of crisis months: the recessionary episodes of 2000–2002 and 2008–2009 account for most of the 1966–2011 out-of-sample gain of the volume rules, and the 2012–2025 period contains no comparable drawdown episode, which mechanically deflates R^2_{OOS}. Second, the post-2011 period is characterized by passive, momentum-dominated flows and rapid arbitrage of published signals, both of which erode return predictability. Third, the joint multivariate fits estimated under ℓ_1/ℓ_2 penalties shrink toward zero and end up overfitting relative to the deliberately simple bivariate design; the Granger–Ramanathan combination, which adapts to past out-of-sample performance (Figure 3), is the closest to the benchmark precisely because it learns to place most weight on the shrinkage forecasts. Consistent with this reading, restricting the predictor matrix to the 84 macro-lag columns improves the joint fit materially (LASSO $R^2_{\text{OOS}} = -0.9\%$ at $h = 1$, versus -8.8% on the full set), while the technical-only variant is also negative (Appendix A): in the multivariate setting the technical signals dilute rather than sharpen the fit. Taken together, our evidence is consistent with Goyal, Welch and Zafirov (2024): the equity premium remains very hard to predict out-of-sample, and any predictability is episodic rather than structural.

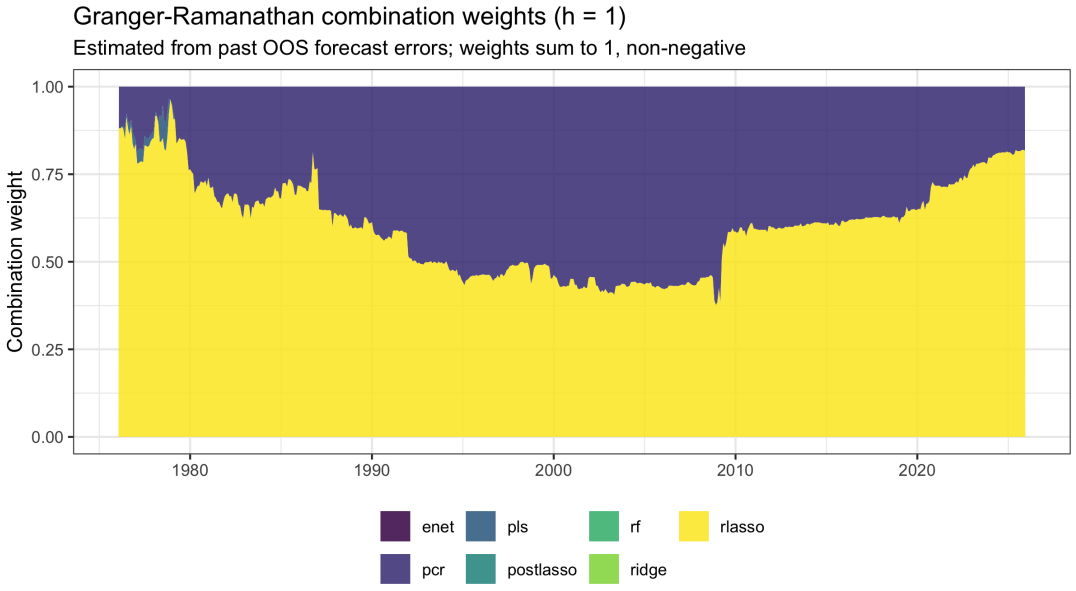


Figure 3: Granger–Ramanathan combination weights ($h = 1$), estimated from past out-of-sample forecast errors with $\sum_i w_i = 1$, $w_i \geq 0$.

5 Conclusion

We replicate and extend Neely et al. (2014) on a 1951–2025 sample with a fixed 240-month rolling design, seven regularized machine learning models and two forecast combinations. The replication validates our pipeline: the computed technical signals match the published series in 99.98% of months, and the volume-based signals are again significantly predictive on the original 1966–2011 evaluation period. On the extended sample, however, no model beats the

prevailing historical mean at any horizon—the premium predictability documented a decade ago has decayed in the post-2011 data. The forecasts nonetheless retain economic value: switching strategies halve the maximum drawdown at market-level Sharpe ratios, and a mean–variance investor gains 1.2–1.3% per annum in certainty-equivalent terms net of transaction costs.

Our study has clear limitations. It covers a single market (the US) with monthly data; the rolling window length and lag structure are fixed rather than selected; and the economic evaluation ignores short-sale constraints beyond the $[0, 1.5]$ weight bound and assumes the investor trades at the index level. Future work could extend the exercise to international equity markets, higher-frequency signals, or shrinkage schemes that combine the prevailing mean directly with the model forecast—a natural response to the persistent failure of point forecasts documented here.

A Predictor-set variants

Table 5 reports R^2_{OOS} at $h = 1$ for three predictor sets: the full design (84 macro-lag columns plus 14 technical signals), the 14 technical signals alone, and the 84 macro-lag columns alone. The macro-only set is closest to the benchmark for the shrinkage estimators (LASSO -0.9%), and the technical-only set is uniformly worse—in the joint multivariate fit the technical signals dilute rather than sharpen the forecasts, which is why the bivariate design of Section 4.2 remains the natural vehicle for their information.

Table 5: Out-of-sample R^2 (%) at $h = 1$ by predictor set: full set (98 columns), technical signals only (14), and macro lags only (84).

Model	Full	Technical	Macro
rlasso	-8.79	-9.53	-0.87
postlasso	-17.93	-17.16	-5.45
ridge	-26.64	-25.96	-2.08
enet	-28.84	-27.87	-2.99
pcr	-15.56	-33.65	-2.71
pls	-40.62	-32.21	-7.39
rf	-48.44	-27.51	-40.87
equal_comb	-18.66	-22.02	-3.15
gr_comb	-10.04	-10.18	-1.63

B Clark–West appendix test

Because every model in this paper nests the prevailing mean (a regression on a constant only), we supplement the Diebold–Mariano tests of Table 2 with the Clark–West (2007) adjusted-MSPE test, which is designed for nested comparisons. The one-sided p-values, reported in Table 2 alongside the DM statistics, confirm the DM verdict: no model significantly improves on the benchmark at any horizon, and the replication exercise of Section 4.2 shows the test has power—it detects the positive 1966–2011 volume-signal predictability at the 5% level. The test is implemented in base R with Newey–West long-run variance (lag $h - 1$); the implementation is cross-checked against the

published results of the original studies in `output/tables/signal_validation.csv` and `output/tables/table5_neely_replication.csv`.

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